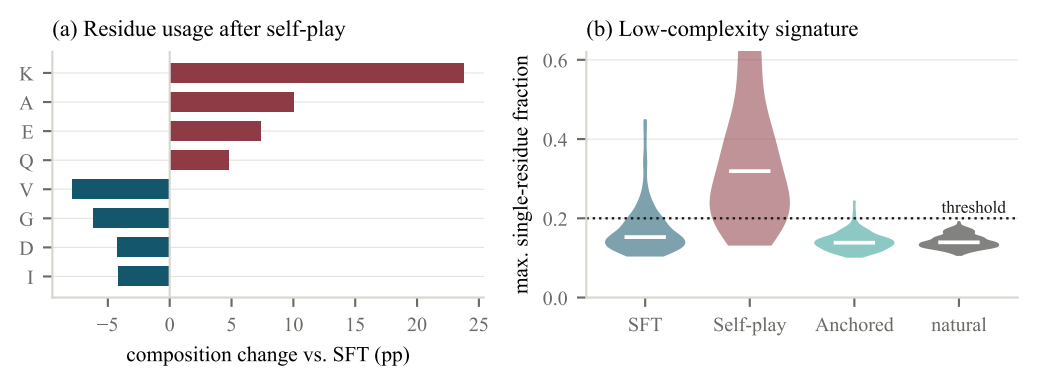
(a) Residue usage after self-play
K
A
E
Q
V
G
D
I
−5
0
5
10
15
20
25
composition change vs. SFT (pp)
(b) Low-complexity signature
max. single-residue fraction
0.6
0.4
0.2
0.0
threshold
SFT
Self-play
Anchored
natural